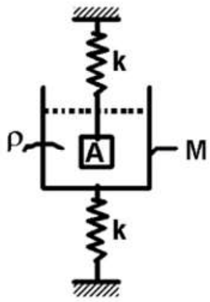
TOPIC : SHM-1

SUBJECT: PHYSICS **DATE: 06-04-2024**

1. The system shown in the adjacent figure is in equilibrium. The mass of the container with liquid is M, density of liquid in the container is ρ and volume of the block is V. If the container is now displaced downwards through a distance x_0 and released such that the block remains well inside the liquid, then during subsequent motion



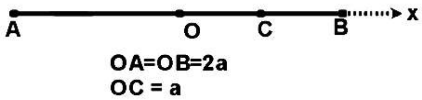
(A) time period of SHM of the container will be $2\pi\sqrt{\dfrac{M}{k}}$

(B) time period of SHM of the container will be $2\pi\sqrt{\dfrac{(M+\rho v)}{k}}$

(C) amplitude of SHM of the container will be greater than x_0

(D) container will not perform SHM

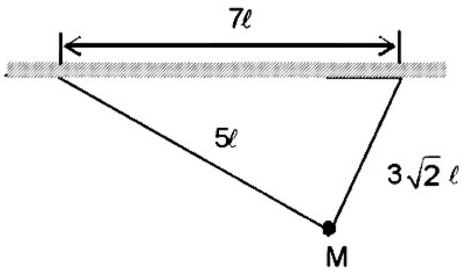
2. A particle of mass m is performing simple harmonic motion along line AB with amplitude $2a$ with centre of oscillation as O. At $\mathrm{t=0}$ particle is at point $\mathrm{C(OC=a)}$ and is moving towards B with velocity $v = a\sqrt{3}\, m/s$. The equation of motion can be given by



(A) $x = 2a(\sin t + \sqrt{3}\cos t)$ (B) $x = 2a(\sqrt{3}\sin t + \cos t)$

(C) $x = a(\sin t + \sqrt{3}\cos t)$ (D) $x = a(\sqrt{3}\sin t + \cos t)$

3. A particle is suspended by two ideal strings as shown in the figure. Now mass m is given a small displacement perpendicular to the plane of triangle formed. Choose the correct statement(s).



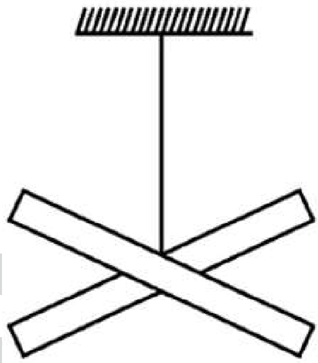
(A) The period of oscillation of the system is $2\pi\sqrt{\dfrac{3\sqrt{3}\ell}{g}}$

(B) The period of oscillation of the system is $2\pi\sqrt{\dfrac{3\ell}{g}}$.

(C) The period of oscillation of the system is independent of M.

(D) If the distance between the suspension points was kept constant and the lengths of the strings were quadrupled then the period of the system will double.

4. A torsional pendulum consists of two identical rods each of mass M & length ℓ joined perpendicular to each other in horizontal plane and is hanged through the wire having torsional constant *C*. The system performs simple harmonic motion $\theta = A\sin(\omega t + \phi)$ such that initially angular displacement of the system is equal to half of amplitude and its angular velocity is ω_0 and is decreasing, then :-



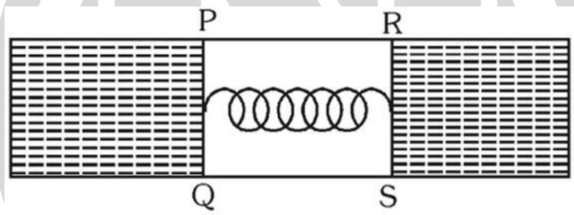
(A) Time period of pendulum is $T = 2\pi\sqrt{\dfrac{M\ell^2}{6C}}$

(B) Amplitude of oscillation is $\dfrac{\omega_0 \ell}{3}\sqrt{\dfrac{2M}{C}}$

(C) Initial phase of SHM is $\dfrac{\pi}{6}$ or $\dfrac{7\pi}{6}$

(D) Energy stored in wire at $t = 0$ is $\dfrac{M\ell^2\omega_0^2}{36}$

5. Two identical straight wires PQ and RS each of mass m and length ℓ can move smoothly on a fixed rectangular frame. Two thin films of a liquid of surface tension T are formed between each wire and the frame. The two wires are connected by a massless spring of stiffness k and intially in a natural length position and released then choose the correct option(s).



(A) Maximum elongation of spring, $\Delta x_m = \dfrac{2T\ell}{k}$

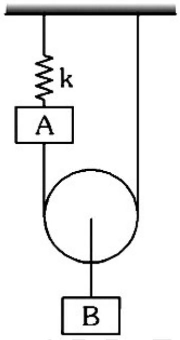
(B) Maximum elongation of spring, $\Delta x_m = \dfrac{4T\ell}{k}$

(C) Each wire executes SHM with the time period, $T_0 = 2\pi\sqrt{\dfrac{m}{2k}}$

(D) Each wire executes SHM with the time period, $T_0 = 2\pi\sqrt{\dfrac{2m}{k}}$

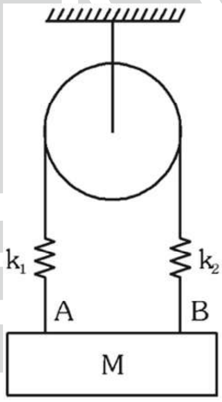
6. In the system shown in figure, Both the blocks A and B have equal mass of m and the force constant of the ideal spring is k. Pulley and threads are massless.

The time period of vertical oscillations of block A is :



(A) $2\pi\sqrt{\frac{3m}{7K}}$ (B) $2\pi\sqrt{\frac{4m}{3K}}$ (C) $2\pi\sqrt{\frac{5m}{4K}}$ (D) $2\pi\sqrt{\frac{5m}{2K}}$

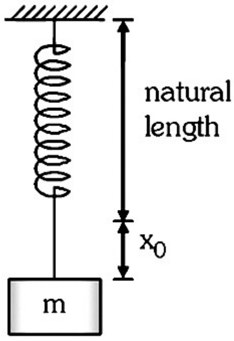
7. In the system shown in the figure the string, springs and pulley are light. The force constant of the two springs are $k_1 = k$ and $k_2 = 2k$. Block of mass M is pulled vertically down from its equilibrium position and released. The top surface of the block (represented by line AB) always remains horizontal. The angular frequency of oscillation is found to be $\sqrt{\frac{CK}{3m}}$. The value of C is



(A) 3 (B) 5 (C) 6 (D) 8

Passage (Q.8 & 9)

A spring block pendulum is shown in the figure. The system is hanging in equilibrium. A bullet of mass $(m/2)$ moving at a speed u hits the block from downward direction and gets embedded into it.



8. The amplitude of oscillation of block is

(A) $\sqrt{\frac{mu^2}{2k}+\left(\frac{mg}{2k}\right)^2}$ (B) $\sqrt{\frac{mu^2}{4k}+\left(\frac{mg}{2k}\right)^2}$ (C) $\sqrt{\frac{mu^2}{6k}+\left(\frac{mg}{2k}\right)^2}$ (D) $\sqrt{\frac{mu^2}{8k}+\left(\frac{mg}{2k}\right)^2}$

9. The time taken by block to reach its upper extreme position after hit by bullet is ($A =$ amplitude)

(A) $\sqrt{\frac{3m}{5k}}\cos^{-1}\left(\frac{mg}{3kA}\right)$ (B) $\sqrt{\frac{3m}{k}}\cos^{-1}\left(\frac{mg}{3kA}\right)$ (C) $\sqrt{\frac{3m}{2k}}\cos^{-1}\left(\frac{mg}{2kA}\right)$ (D) $\sqrt{\frac{m}{2k}}\cos^{-1}\left(\frac{mg}{3kA}\right)$

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ANSWER KEY

Ques.	1	2	3	4	5	6	7	8	9
Ans.	B	D	ABCD	BC	BC	C	D	C	C